

Class-Aware Prompted K-BERT for Advertisement Compliance Detection

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Abstract—Advertisement compliance detection plays an important role in regulating online commercial content. Although pre-trained language models and knowledge-enhanced methods such as K-BERT have achieved strong performance in text classification, existing approaches usually rely on uniform knowledge injection and lack task-oriented discriminative guidance. In advertisement compliance detection, however, violation identification is often dominated by surface-level lexical cues, making effective decision boundary modeling particularly important. To address this issue, we propose a Class-Aware Prompted K-BERT framework that incorporates category-specific soft prompts into the K-BERT architecture. The proposed method guides the model to interpret advertisement texts from a compliance-aware perspective and enhances category-sensitive representation learning. In addition, we redesign the visible matrix to support effective interaction between prompt tokens and textual representations while preserving the hierarchical structure of knowledge injection. Experiments on an OCR-based e-commerce advertisement dataset show that the proposed method outperforms both BERT and K-BERT. The ablation study further indicates that direct knowledge injection does not always guarantee performance gains, while the proposed class-aware prompt mechanism can effectively improve advertisement compliance detection.

Keywords—advertisement compliance detection; K-BERT; class-aware prompt learning; knowledge-enhanced language model; e-commerce advertising

I. INTRODUCTION

The rapid expansion of e-commerce platforms has fundamentally transformed the landscape of commercial promotion. Online advertisements have become a primary channel for marketing communication, influencing purchasing decisions at an unprecedented scale. However, this growth has also led to a proliferation of non-compliant advertising content, including exaggerated claims, unverifiable efficacy statements, and misleading guarantees. Such content not only undermines consumer trust but also poses significant challenges for

regulatory enforcement. Recent studies have explored fact-checking for online advertisement posts^[1], regulatory compliance analysis in social media influencer marketing^[2], and sensitive content classification in broader moderation settings^[3]. Consequently, automated advertisement compliance detection has emerged as an important task in intelligent content moderation systems.

Recent advances in pre-trained language models (PLMs) have significantly improved the performance of various text classification tasks. Models such as BERT^[4] leverage large-scale pre-training to capture contextualized semantic representations, enabling strong downstream performance. To further enhance semantic understanding, knowledge-enhanced models, most notably K-BERT^[5], incorporate structured knowledge graphs into the encoding process. By injecting external knowledge into the input sequence and regulating attention through a visible matrix, K-BERT enriches contextual representations and improves reasoning ability in tasks involving implicit semantic relations.

Despite these advantages, the assumption that knowledge injection universally benefits classification performance deserves reconsideration in domain-specific scenarios. Advertisement compliance detection differs from many conventional language understanding tasks. In practice, violation identification is often driven by explicit surface-level lexical cues, such as absolute expressions (“best”, “guaranteed”), exaggerated promotional claims, or implied medical effects. These signals frequently serve as strong discriminative indicators that can be captured directly from the original text. In such cases, additional knowledge may not necessarily improve classification performance and may even introduce redundant semantic information, especially when the external knowledge is not fully aligned with the target compliance scenario^{[1], [2]}.

Motivated by this observation, we propose a Class-Aware Prompted K-BERT framework for advertisement compliance detection. Instead of modifying the encoder architecture, the proposed method introduces label-specific soft prompt

embeddings to guide the model toward compliance-oriented representation learning. By associating different prompts with different class hypotheses, the model interprets the same advertisement text from multiple decision perspectives, effectively reshaping the classification space. Our design is inspired by recent advances in prompt-based adaptation and prompt optimization^{[6]-[8]}. In addition, we extend the visible matrix mechanism to ensure coherent interaction between prompt tokens and textual representations while maintaining the hierarchical structure of knowledge injection.

Extensive experiments on an OCR-based e-commerce advertisement dataset demonstrate that the proposed framework consistently outperforms the original K-BERT implementation. Interestingly, strong performance is preserved even without knowledge graph injection, highlighting the dominant role of discriminative prompt guidance in surface-trigger-dominated compliance scenarios. These findings provide new insights into the interaction between prompt learning^{[6]-[8]} and knowledge-enhanced architectures^[5].

II. RELATED WORK

A. Advertisement Compliance Detection

Early studies on advertisement compliance detection mainly relied on rule-based filtering and manually designed keyword lists. Although such approaches can capture explicit violation patterns, they often lack robustness when dealing with diverse linguistic expressions. With the development of deep learning, neural text classification models based on CNNs, RNNs, and transformer architectures have been widely adopted for content moderation tasks. In particular, pre-trained language models (PLMs) such as BERT^[4] have significantly improved performance by leveraging large-scale contextual pre-training. Recent studies have also explored online advertisement fact-checking, regulatory compliance analysis, and sensitive content moderation in related settings^{[1]-[3]}. However, most existing methods treat compliance detection as a standard classification problem and rarely consider task-specific modeling characteristics.

B. Knowledge-Enhanced Language Models

To enhance semantic representation, knowledge-enhanced models incorporate structured knowledge graphs into language models. K-BERT^[5] extends BERT by injecting knowledge triples into the input sequence and controlling attention through a visible matrix mechanism. Such methods have shown advantages in reasoning and relation extraction tasks. Nevertheless, existing knowledge injection strategies typically apply external knowledge uniformly to all inputs, without considering whether knowledge augmentation is necessary for specific classification scenarios.

C. Prompt Learning in Pre-trained Language Models

Prompt learning^[6] has recently emerged as an effective paradigm for adapting pre-trained language models to downstream tasks. Soft prompt tuning introduces learnable continuous embeddings into the input sequence, enabling efficient task adaptation with limited additional parameters.

Prompt mechanisms have been shown to reshape decision boundaries by guiding the model toward task-relevant features. However, the integration of class-aware prompt design with knowledge-enhanced architectures remains underexplored, particularly in domain-specific compliance detection tasks.

III. METHOD

A. Problem Definition

Given an advertisement text $X = \{x_1, x_2, \dots, x_n\}$, the task of advertisement compliance detection aims to predict its compliance label $y \in \{0, 1\}$, where 0 denotes compliant content and 1 denotes non-compliant content. The objective is to learn a classification function:

$$f: X \rightarrow y \quad (1)$$

Unlike general text classification tasks that rely on implicit semantic reasoning, compliance detection in advertising scenarios is often dominated by explicit surface-level lexical patterns, such as exaggerated claims and absolute expressions. These features act as strong discriminative signals, suggesting that modeling the decision boundary effectively may be more critical than enriching semantic representations through external knowledge.

B. Revisiting K-BERT

K-BERT extends the standard BERT architecture by injecting knowledge graph triples into the input sequence. For each token associated with an entity, relevant knowledge triples are expanded into a tree-like structure. The augmented input can be represented as:

$$H = [CLS, X, K] \quad (2)$$

where K denotes the knowledge-injected tokens.

To prevent semantic drift introduced by knowledge expansion, K-BERT employs a visible matrix M to restrict attention propagation. Specifically:

- Original text tokens attend to each other globally.
- Knowledge tokens attend only to their anchor tokens and hierarchical paths.
- Cross-branch attention is constrained.

Although this mechanism enhances contextual representation, the knowledge injection process is uniformly applied across all samples, implicitly assuming that additional knowledge consistently benefits classification performance. Based on the original K-BERT backbone, our method introduces class-aware prompt embeddings and extends the visible matrix mechanism to better support compliance-oriented representation learning. The overall architecture of the proposed framework is shown in Fig. 1.

C. Motivation for Class-Aware Prompting

In advertisement compliance detection, violation signals are frequently explicit rather than implicit. Therefore, injecting additional semantic knowledge may not always improve

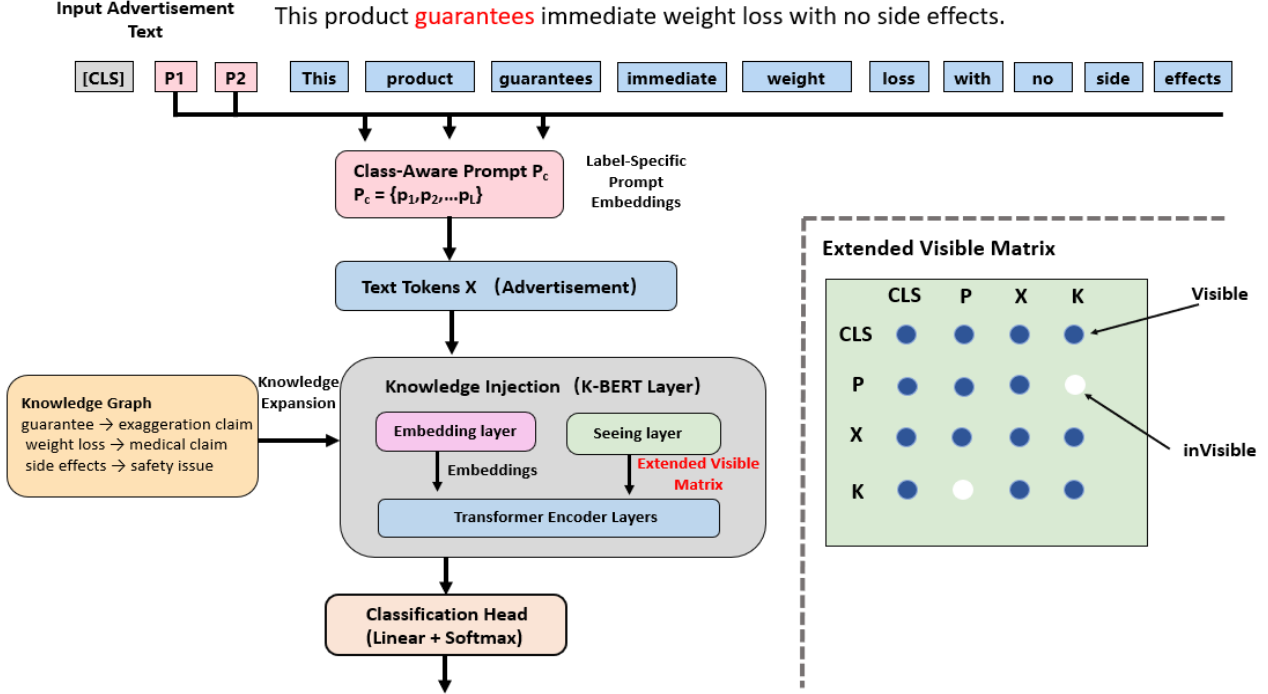


Figure 1. Architecture of the proposed Class-Aware Prompted K-BERT framework for advertisement compliance detection.

performance. Instead, guiding the model to interpret text from a task-specific perspective can be more beneficial.

Let the encoder output representation under class hypothesis c be:

$$h_c = \text{Encoder}(X; \theta, P_c) \quad (3)$$

Where P_c denotes a class-specific prompt embedding. By associating distinct prompts with different labels, the model is encouraged to construct multiple decision hypotheses over the same input text.

This mechanism effectively reshapes the classification boundary:

$$y = \arg \max_{c \in \{0,1\}} f(h_c) \quad (4)$$

Instead of expanding semantic information through knowledge injection, the proposed class-aware prompts guide the model to construct multiple classification hypotheses for the same input.

The class-aware prompt mechanism can be interpreted as constructing multiple classification hypotheses for the same input text. As illustrated in Fig. 2, different prompt embeddings guide the encoder to produce class-specific representations.

D. Class-Aware Prompt Design

Inspired by recent advances in prompt learning and prompt optimization^{[6]-[8]}, we introduce class-specific prompt embeddings into the K-BERT framework. Unlike generic soft prompts, our prompt embeddings are explicitly tied to class hypotheses for compliance-oriented decision modeling. For a

binary classification task, we define two learnable prompt embedding sets:

$$P_c = \{p_{c,1}, p_{c,2}, \dots, p_{c,L}\}, c \in \{0,1\} \quad (5)$$

Each prompt embedding $p_{c,i} \in \mathbb{R}^d$ shares the same dimensionality as token embeddings. These embeddings are initialized randomly and optimized jointly with the downstream objective.

For each class hypothesis c , the final input sequence is constructed as:

$$H_c = [[CLS], P_c, X, K] \quad (6)$$

Prompt tokens are inserted immediately after the [CLS] token to ensure early interaction with contextual representations.

E. Visible Matrix Extension

To preserve the structural constraints of K-BERT while enabling prompt interaction, we extend the visible matrix M as follows:

1. Prompt tokens are globally visible to [CLS] and original text tokens.
2. Knowledge tokens remain restricted to their corresponding anchor tokens.
3. Prompt tokens do not directly attend to lower-level knowledge nodes.

This design ensures that prompt embeddings function as global discriminative controllers without disrupting hierarchical knowledge structure.

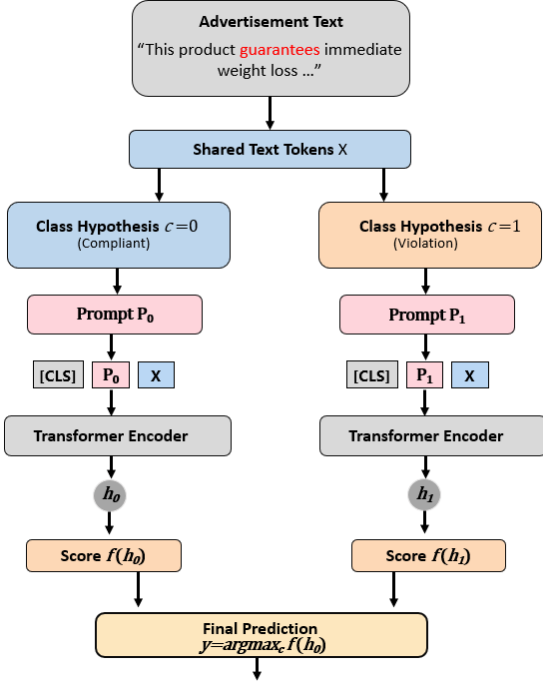


Figure 2. Illustration of the class-aware prompt hypothesis mechanism. Different prompt embeddings generate class-specific representations for the same advertisement text.

F. Training and Inference Strategy

During training, only the prompt corresponding to the ground-truth label is activated. The classification probability is computed as:

$$\hat{y} = \text{softmax}(W h_{\text{CLS}}) \quad (7)$$

The model is optimized using cross-entropy loss:

$$\mathcal{L} = - \sum y \log \hat{y} \quad (8)$$

During inference, we adopt a prompt-enumeration strategy:

$$s_c = f(H_c) \quad (9)$$

The final prediction is determined by selecting the class with the highest score.

G. Complexity Analysis

The additional parameters introduced by class-aware prompts are:

$$\Delta P = C \times L \times d \quad (10)$$

Where C is the number of classes, L is the prompt length, and d is the embedding dimension.

Compared with the full parameter size of BERT-based models, the additional prompt parameters are negligible. The computational complexity during inference increases linearly

with the number of classes due to prompt enumeration, which remains efficient for binary classification tasks.

IV. EXPERIMENTS

A. Dataset and Experimental Settings

To evaluate the proposed method, experiments are conducted on an OCR-based e-commerce advertisement dataset. The dataset is constructed by collecting advertisement images from e-commerce platforms and extracting textual content through optical character recognition (OCR). It contains approximately 12.4k samples, with about 7.8k labeled as 0 and about 4.6k labeled as 1, where label 1 denotes non-compliant advertisements. The violations mainly involve exaggerated claims, unverifiable efficacy descriptions, and misleading guarantees. The dataset is divided into training, development, and test sets with a ratio of 80%/10%/10%. For fair comparison, all models were trained and evaluated under the same experimental settings. Specifically, the maximum sequence length was set to 256, the batch size was set to 32, the learning rate was set to $2e-5$, and the number of training epochs was set to 10. The encoder type was set to BERT, and the first-token pooling strategy was adopted for sentence representation. In addition, the random seed was fixed at 7 to improve reproducibility.

B. Compared Methods

To verify the contribution of each component, we compared the following three models:

BERT: a vanilla BERT-based text classification model without knowledge injection or prompt enhancement. It serves as the basic pretrained language model baseline.

K-BERT: extends BERT by injecting structured knowledge graphs into the input sequence and regulating attention through a visible matrix mechanism.

Class-Aware Prompted K-BERT: the proposed method, which extends K-BERT by introducing class-aware soft prompts. Specifically, class-dependent prompt embeddings are incorporated into the input representation so that the model can better capture category-sensitive semantic patterns for advertisement compliance detection.

C. Overall Performance Comparison

To evaluate the overall effectiveness of the proposed method, we compare BERT, K-BERT, and the proposed Class-Aware Prompted K-BERT on the advertisement compliance detection dataset. The results are reported in Table I, and Fig. 3 further illustrates the comparison in terms of Accuracy and F1-score.

As shown in Table I, the proposed Class-Aware Prompted K-BERT achieves the best performance among all compared methods. Compared with BERT and K-BERT, the proposed method obtains higher Precision, Recall, F1-score, and Accuracy, demonstrating the effectiveness of introducing class-aware prompt learning into the knowledge-enhanced framework.

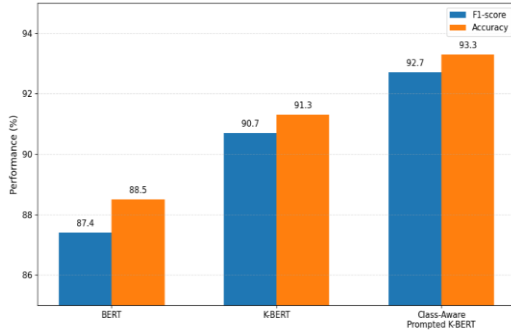


Figure 3. Performance comparison of different models on the advertisement compliance detection dataset in terms of Accuracy and F1-score.

This improvement indicates that the class-aware prompt mechanism can effectively enhance category-sensitive representation learning and help the model better capture discriminative semantic cues for compliance judgment. As a result, the proposed method shows stronger ability in distinguishing compliant and non-compliant advertisements.

TABLE I. PERFORMANCE COMPARISON BETWEEN DIFFERENT MODELS ON THE ADVERTISEMENT COMPLIANCE DETECTION DATASET.

Model	Precision	Recall	F1	Accuracy
BERT	87.6	87.7	87.4	88.5
K-BERT	90.4	91.0	90.7	91.3
Class-Aware Prompted K-BERT (Ours)	93.0	92.6	92.7	93.3

The best results are highlighted in bold.

D. Ablation Study

To further analyze the effectiveness of different components, we perform an ablation study with three variants: K-BERT w/o KG, K-BERT, and Class-Aware Prompted K-BERT. The results are reported in Table II.

Interestingly, K-BERT w/o KG performs slightly better than the original K-BERT, suggesting that direct knowledge injection does not always lead to better performance. One possible reason is that the injected knowledge may contain noisy or task-irrelevant information, which interferes with the original textual semantics. In this study, the domain-specific knowledge graph was mainly constructed from advertising regulations. Candidate triples were manually extracted from legal provisions and compliance-related rules, such as “cure completely”-belongs to-“absolute expression”, and further reviewed by relevant experts to ensure legal validity and compliance relevance. Although this process improves knowledge quality, the current graph may still have limited coverage and adaptability, which may partly explain the noise introduced by direct knowledge injection.

The proposed Class-Aware Prompted K-BERT achieves the best results on all metrics, demonstrating that the class-aware prompt mechanism can effectively enhance category-sensitive representation learning and improve the utilization of knowledge-enhanced features. These results confirm that the proposed design provides a more effective framework for advertisement compliance detection.

TABLE II. ABLATION STUDY OF DIFFERENT COMPONENTS ON THE ADVERTISEMENT COMPLIANCE DETECTION DATASET.

Model	Precision	Recall	F1	Accuracy
K-BERT w/o KG	92.2	91.6	91.9	92.5
K-BERT	90.4	91.0	90.7	91.3
Class-Aware Prompted K-BERT (Ours)	93.0	92.6	92.7	93.3

The best results are highlighted in bold.

V. CONCLUSIONS

In this paper, we proposed a Class-Aware Prompted K-BERT framework for advertisement compliance detection. By introducing class-aware prompt learning into K-BERT, the proposed method enhances category-sensitive representation learning and improves the discrimination between compliant and non-compliant advertisements.

Experimental results show that the proposed method outperforms both BERT and K-BERT. The ablation study further indicates that direct knowledge injection does not always guarantee performance gains, while the proposed class-aware prompt mechanism effectively improves classification performance. In future work, we will explore higher-quality advertisement knowledge graphs and more effective prompt learning strategies.

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